

# Reading: Intellectual Property and Types of Confidential Information

Estimated time needed: 5 minutes

An information asset is information or data that is of value. Examples include patient records, customer information, and intellectual property. Information assets can exist physically, on paper, disks, in other media, or electronically in databases and files.

- Data analytics is when raw data, like values or facts, create meaningful information. Data is the raw values and facts, usually collected by automated systems. For example, page visits, link clicks, and monthly sales.
- Information is a summary of the raw data. For example, positive or negative results that happen after some specific change.
- Insights are conclusions based on the results of information analysis. Meaningful business decisions are based on insights. For example, if a positive trend occurs after a manager changes store hours, the correct business decision would be to maintain those new hours.
- Intellectual property (or IP) refers to creations of the mind and generally are not tangible. Copyright, trademark, and patent law often protect IP. Some IP examples are industrial designs, trade secrets, and research discoveries. Even some employee knowledge is considered intellectual property. Companies use a legally binding document called a Non-Disclosure Agreement (or an NDA) to prevent sharing of sensitive information.

## Digital Products

Digital products are non-tangible assets a company owns. Examples include software, online music, online courses, e-books or audiobooks, and web elements like WordPress or Shopify themes. A company must protect digital products from piracy and reverse engineering. Source codes, licenses, and activation keys also need protection from hackers and insider threats.

- Digital Rights Management, or DRM, is code added directly to files that helps prevent digital assets from being copied or pirated, but some tools can remove DRM code.
- The Digital Millennium Copyright Act, or DMCA, makes it illegal to bypass copy protections or to develop technology that helps bypass copy protections.

Data-driven business decisions help companies respond to real events. For example, sales and marketing data helps identify trends and customer interests. And production and fulfillment data help identify productivity issues in manufacturing, billing systems, transportation, and more. Getting the correct information is key to data-driven business decisions.

- Data capture is the collection of data from multiple sources and storing the secure storage of it securely in relational databases or, more commonly, semi-structured data warehouses. Data may be captured by: Server logs showing where customers browse, IoT sensors in home appliances and business technology, customer and employee surveys, or rating systems.
- Data correlation is when raw data points are analyzed to find connections or links. For example, Netflix uses tools that compare searches, views, and ratings to predict which movies and shows will be successful on their platform. AI and machine learning algorithms automate parts of the analysis.
- Meaningful reporting is the presentation of analyzed information in ways that help people further analyze and interpret. Reporting tools use captured and correlated data to provide charts, keyword searches, and graphs that help companies achieve business insights.

## Confidential Information

Confidential information is information that employees must keep secret. Employees are trained to recognize and deal with confidential information so that it remains secure. Companies rank information and files by how sensitive each one is. Each company ranks its information differently, but there are four main types of confidential information that should be universally protected:

- Personally Identifiable Information (PII) is any information that can be used to identify someone, like government ID numbers, birthdates, addresses, and phone numbers.
- Company Confidential information is any information used to run a company, like intellectual property, product designs, procedures, plans, employee records, and financial data.
- Customer Confidential Information is information customers or partners provide to companies, which includes PII and things like purchase histories and credit card information.
- Protected Health Information (PHI) is any information added to a person's medical record during diagnosis or treatment that can be used to identify them, like PII, medical history, prescription lists, photos, and more. Examples of careless data handling include entering a customer's credit card information into an unencrypted database, leaving a patient's medical file unattended at the front desk, or letting a work friend borrow your password to download files because they forgot theirs.

Some examples of properly handling confidentiality include:

- Restricting access to only those who need the information
- Not allowing unauthorized views or copies to be created
- Storing information securely with encryption, firewalls, permissions, and more
- Destroying any file copies that are no longer needed—not just discarding them
- Getting explicit consent before processing or storing information, including disclosure about how long the company will keep it
- Ensuring employees create strong passwords that they do not write down or share and that they change those passwords regularly.

## PII, PCI, and SPI

PII is sometimes confused with PCI and SPI because they're very similar. In most non-legal situations, these terms are often used interchangeably. Here are the differences between them.

- Personally Identifiable Information (PII) is information that identifies a person.
- Personal Customer Information (PCI) identifies and describes a customer. It includes many of the same data types as PII, like name, address, contact information, account login, and demographics. It can also include descriptive data like age, gender, job title, and marital status.
- Sensitive Personal Information (SPI) is information that does not identify but can cause harm if made public.



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